



AI Agents at Scale: A data scientist's journey to transform clients with tech



QuantumBlack | Tech & AI | Our People

October 17, 2025 - As a [principal data scientist](#) at QuantumBlack, AI by McKinsey, I build AI solutions that solve business problems for clients. I focus on our [Agents at Scale](#), a factory of AI agents working together to take on large, complex digital transformations.

Here's how I landed in this field, a look at the complexity of AI agents working together, and what the fast pace of change means for the future.



Why machine learning and agents spark my curiosity

When I started my undergraduate degree in mathematics, I was convinced my future lay in finance. During my internship at an investment bank, I searched for a team that used data to predict future outcomes. To my surprise, I didn't find what I was looking for.

Out of curiosity, I picked up a book on the subject during my final year at university—and I was hooked. Suddenly, I saw how data could be used to predict, simulate, and solve real-world problems and pursued a master's degree in data science and machine learning.

When I discovered QuantumBlack at a conference in London, its vision of applying advanced analytics to complex business challenges resonated with me, and I joined soon after completing my studies.

At my core, I'm analytical. I like hard facts, numbers, and exploring all possible pathways before making a decision. Machine learning is a natural extension of that mindset—it's about using data to understand how the world works and to tackle big problems elegantly. Over the years, my career has taken me across industries and applications, but I've always been drawn to challenges without obvious solutions. That curiosity eventually led me to the Agents at Scale work we're pioneering at McKinsey.

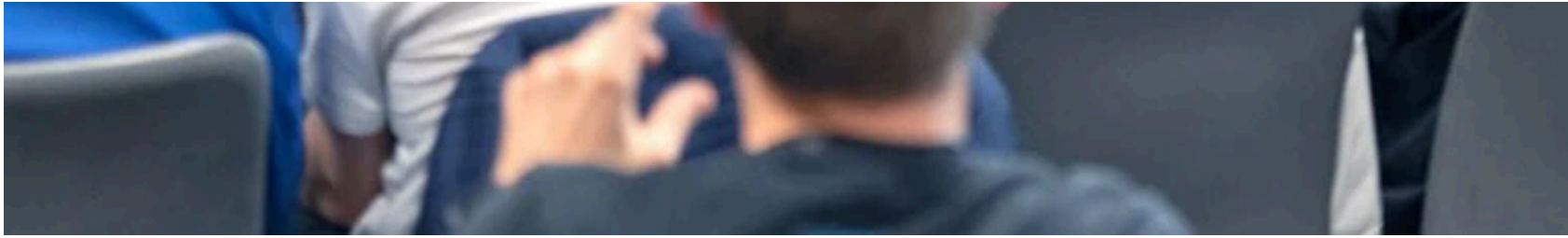
Building Agents at Scale

The slide is divided into two columns. The left column, titled '3. Practitioner using Agent Assistants', features an icon of a person and a robot. It states that practitioners or events invoke agents to create outputs or perform tasks, with a '2X' increase in raw productivity potential. The right column, titled '4. Practitioners supervising an Agent Digital Factory', features an icon of a person supervising multiple agents. It states that practitioners build and supervise a virtual organization of agents, with a '20X' increase in raw productivity potential (1 human for 15-20 agents). A bracket on the left indicates that the observed activity is 2x at constant quality in early stages. The footer reads 'QuantumBlack, AI by McKinsey'.

Practitioner using Agent Assistants	Practitioners supervising an Agent Digital Factory
Practitioners or events invoke Agents which create outputs or perform a task e2e	Practitioners build and supervise a virtual organization of agents. If needed, humans finalize outputs
2X Raw productivity potential	20X Raw productivity potential 1 Human for 15-20 agents

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When I joined the team working on what we called LegacyX, our goal was to use generative AI to modernize legacy technology. That project became the foundation for what is now Agents at Scale. It brings various agents, which are self-directed AI systems used for specific goals, together to work in teams to automate large, complex processes.

I've had the privilege of being part of the journey from the beginning, watching the work evolve from early experiments to the deployment of sophisticated, specialized agentic AI workforces.

One project I'm especially proud of involved modernizing a client's legacy Common Business-Oriented Language (COBOL) mainframe. Their system had been in use for more than 20 years, and very few people understood how it worked. We took on the challenge of re-architecting 15,000 lines of COBOL into under 1,000 lines of modern Java. The transformation wasn't about line-by-line translation; it was about rethinking the entire architecture of the system. That's where agents came in.

“This work goes far beyond building technology—it’s about unlocking value for clients and their people.”

— Emily Sagar-Jones

A factory of agents at work

I often describe this process as a digital factory floor—done by agents we'd built. On one side, agents combed through COBOL code, decoding decades of business logic. Their work was digitally handed to architect agents, who reimagined the design in modern, modular patterns, and flagged questions where human validation was needed. Coder agents then brought those designs to life in Java, producing clean, efficient, and maintainable code. At the far end of the floor, tester and fixer agents ran the new code in a sandbox, automatically detecting and correcting errors, much like a quality assurance team working continuously without breaks.

The handoffs between these agents were seamless. Each one had a clear role, but the real power came from how they built on each other's outputs. To make this possible, we integrated tools, enabled by the Model Context Protocol (MCP) system, which allowed tester agents to execute, validate, and repair code autonomously. Debugging—traditionally a painstaking process—became dramatically faster and more scalable.

We achieved over 90 percent accuracy, meaning less than 10 percent of the code required human correction to achieve functional equivalency. Even more striking, the process was twice as fast as manual modernization, even though we were still building and refining the agents as we worked. And, by simply giving the agents different instructions and roles and building a slightly different factory, we can apply them to many other processes, such as identity verification.

Bridging tech and business





This work goes far beyond building technology—it's about unlocking value for clients and their people. A major part of our role is bridging the gap between cutting-edge technology and industry expertise. Many clients have little prior exposure to agentic systems, so we focus on designing processes with them, validating outputs together, and helping them build the capabilities to sustain the work long after we leave.

There's tension here with companies' in-house coders, who want to keep coding. But their jobs as coders don't go away—what they code will change, and a lot of tedious tasks will be automated.

These are still early days for agentic systems. Will we evolve toward single “master agents” that can spin up specialized teams as needed? How can we improve user experience and build greater trust in these systems? And how do we ensure they remain safe, unbiased, and interoperable?

A year ago, I was only experimenting with agents. Today, I'm deploying them with clients and helping them reimagine their operating models. The pace of change is exhilarating, and I'm optimistic about how this work will reshape how organizations function and what we can achieve together.

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Anna Wintour on instinct, inclusion, and bold leadership

Leadership voices

November 24, 2025 - At McKinsey's first *Brilliant Moves* event, Anna Wintour, global editorial director of *Vogue* and chief content officer of Condé Nast, reflected on how the best leaders stay connected: to people, to culture, and to change. Her stories of courage, intuition, and reinvention showed that foresight isn't about prediction; it's about intentional proximity. Her perspective mirrors McKinsey's own focus for its second century: pairing analytical precision with cultural intuition, staying close to clients and communities, and translating insight into impact.

Staying plugged into the pulse of real life

Brilliant moves: Live with Anna Wintour

Anna's early career at *New York Magazine* cemented her belief that great editors must stay immersed in real life. "Working at *New York Magazine* was thrilling," she says. "I was surrounded by people who had absolutely no interest in fashion—and that was wonderful. They made me think about art, film, theater, politics. It made me realize that fashion only matters when it's connected to the world."

That ethos has guided her leadership at *Vogue* for nearly four decades. Even now, she told the audience, she makes time every morning to read multiple newspapers, walk through Washington Square Park, and

call colleagues across time zones before sunrise. “It’s really important to expose yourself to life,” she says. “Walk in the street and take in what you see. It will always give you an idea.”

The power of instinct over analytics

Despite leading one of the world’s most data-literate publishing houses, Anna says she’s never been ruled by metrics. “Forgive me—I know this is McKinsey—but I am not someone who follows data and analytics religiously,” she says with a grin. “I always make my choices with instinct.”

Her most iconic decisions—like *Vogue*’s 1988 debut cover featuring a couture jacket paired with \$50 Guess jeans—were the result of gut feeling, not focus groups. “Everyone was expecting something different,” she remembered. “This image related to the women I saw in the street. I didn’t want *Vogue* to be removed or distant.” The printer even called to ask if the photo had been sent by mistake. That cover went on to redefine modern fashion photography.



Yael Taqqu (left) and Anna Wintour

She recalled a 1989 conversation with a conservative businessman on a plane who said he could never imagine such a thing. “So, I went straight back to the office and said, ‘We’re putting Madonna on the cover,’” she says, laughing. The lesson, she added, was not only about trusting instinct, but also about “engaging people you wouldn’t normally talk to—you never know what you might learn.”

Turning moments into institutions

Anna's creative instincts have done more than sell magazines—they've built cultural institutions. Take the Met Gala. When she was first asked to take the helm of the fundraiser in the 1990s, she thought she'd do it for a year.

Instead, she reinvented it as a global phenomenon. “It didn’t happen overnight,” she explained. “It was a great lesson in patience and listening.”

Anna's stories about later Galas captured both her audacity and her collaborative spirit. She described those moments not as celebrity triumphs but as creative partnerships born of mutual respect and cultural curiosity. "Each exhibition is an adventure," she says. "And it's become truly global. Our last livestream was watched by more than 65 million people."

Inclusion and mentorship as creative engines

For all her reputation as a demanding perfectionist, Anna spoke passionately about inclusion and mentorship. “I like to think of Condé Nast as a place where you can come to learn,” she says. She personally mentors younger talent, adding, “I learn as much from them as they do from me.”

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— Anna Wintour

Her approach to hiring, she explained, is rooted in humanity. “It doesn’t matter what’s on their résumé or where they went to college,” she says. “I ask myself, am I going to be pleased to see this person when they walk into my office?”

That philosophy has shaped the careers of countless creatives. Naomi Campbell, whom she featured on the cover of *Vogue*’s landmark September issue in 1989, at a time when representation on fashion covers was still rare, is just one example. She has ushered in a new generation of designers and editors. “Any great designer has a very strong point of view,” she says. “I can’t sew or make anything. I’m useless with my hands. My job is to put creative geniuses together and support them.”

She sees leadership not as solitary vision, but as enabling others to thrive. Her work founding the CFDA/Vogue Fashion Fund after 9/11 and later transforming it into A Common Thread during the COVID-19 pandemic exemplifies that. “When things are difficult,” she says, “being helpful—using your platform to do good for others—is so rewarding.”

Staying grounded in a life of momentum

Anna is famous for her precision and discipline, and she describes her daily rhythm as a grounding force. “I wake up at 4:30, go to the gym, walk through the park, and read,” she says. “But I also believe in

balance—leaving at a reasonable hour, spending time with family and friends.” Her weekends in the country, she says, remain sacred. “I need quiet time. I find nature very sustaining.”

Even her reflections on resilience returned to that grounded optimism and focus on hope. During COVID-19, she says, “We had to nurture everyone at Condé Nast, help them find ways to create from home, and keep the spirit alive.” She found solace in helping others and in one simple image: a flower photograph by Irving Penn she chose for *Vogue*’s cover. “It just says to me hope, and the future, and beauty,” she says. “It’s honestly my favorite cover I’ve ever published.”

The art of leading with curiosity

Anna’s conversation ended where it began—with curiosity. Whether recalling advice from Ralph Lauren (“You don’t want to be too hot, you don’t want to be too cold, but you always want to be there”) or describing how walking through the park fuels her ideas, she emphasized that great editorial leadership starts with engagement—with the world, with people, with culture in motion.

For Anna, instinct, inclusion, and mentorship aren’t just values—they’re disciplines. They’re how she keeps *Vogue* both timeless and timely. “You have to change and move with the times,” she says, “but you can’t go in the wrong direction. You have to stay true to your own point of view.”

That, perhaps, is Anna Wintour’s most brilliant move of all.